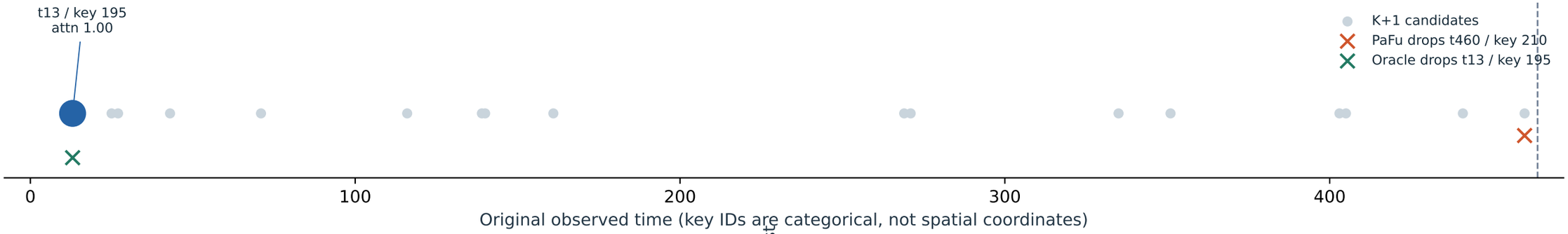
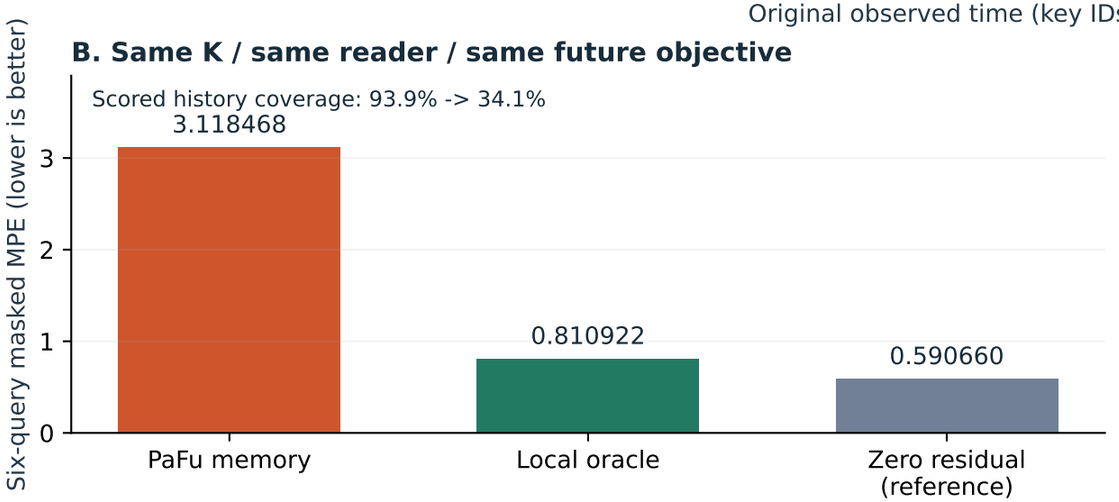
01 | Attention selects a harmful old value
CHANGE / source 31 / K16/PaFu / t=464

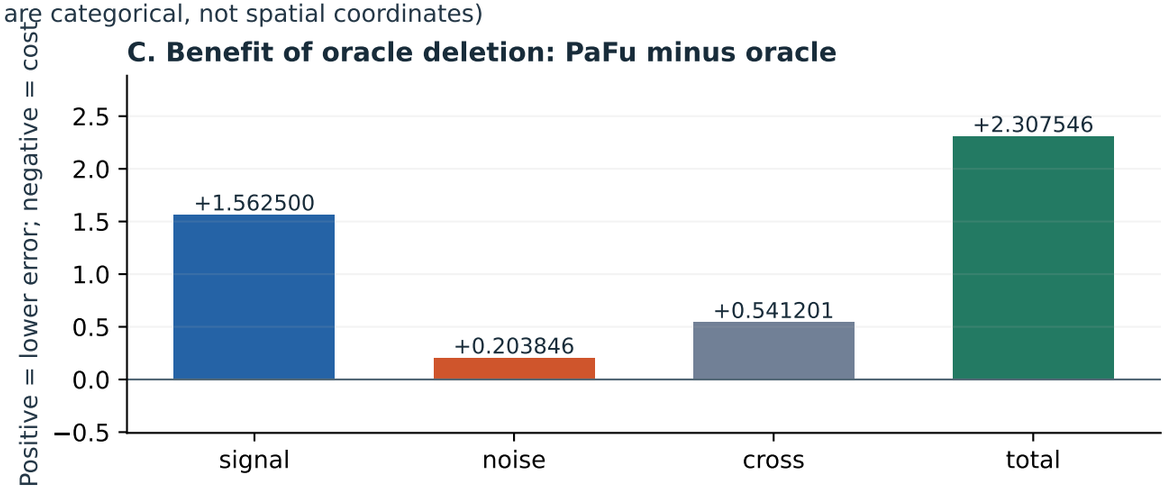
A. Observed candidate history and archived future-attention labels



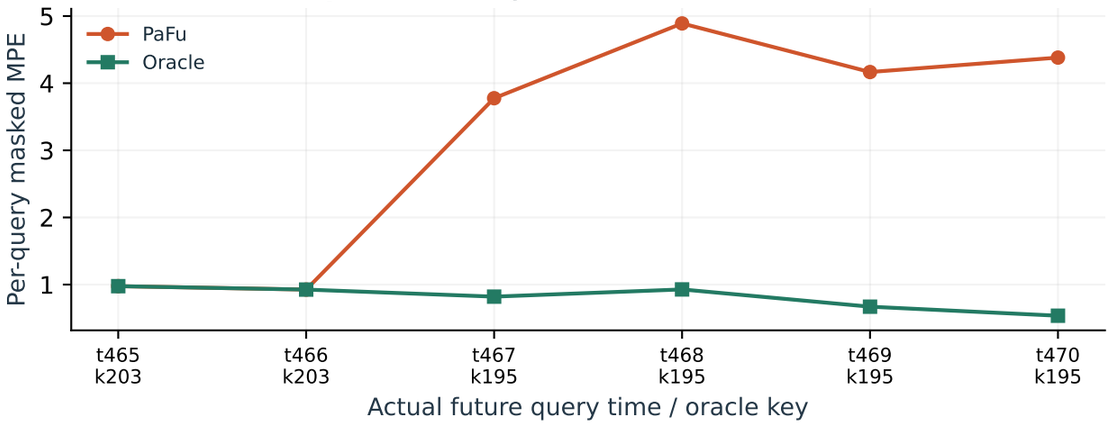
B. Same K / same reader / same future objective



C. Benefit of oracle deletion: PaFu minus oracle



D. Which future queries change?



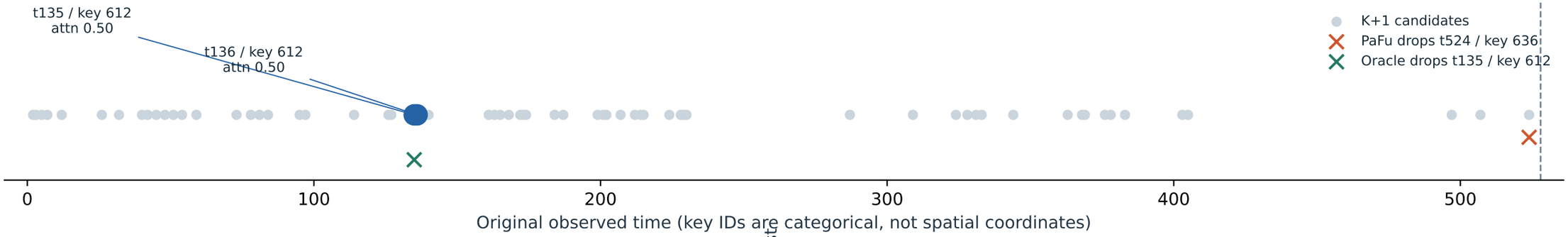
E. Covered scored coordinates / total scored

time / key	PaFu	Oracle
465 / 203	14/14	14/14
466 / 203	14/15	14/15
467 / 195	12/13	0/13
468 / 195	12/13	0/13
469 / 195	12/13	0/13
470 / 195	13/14	0/14

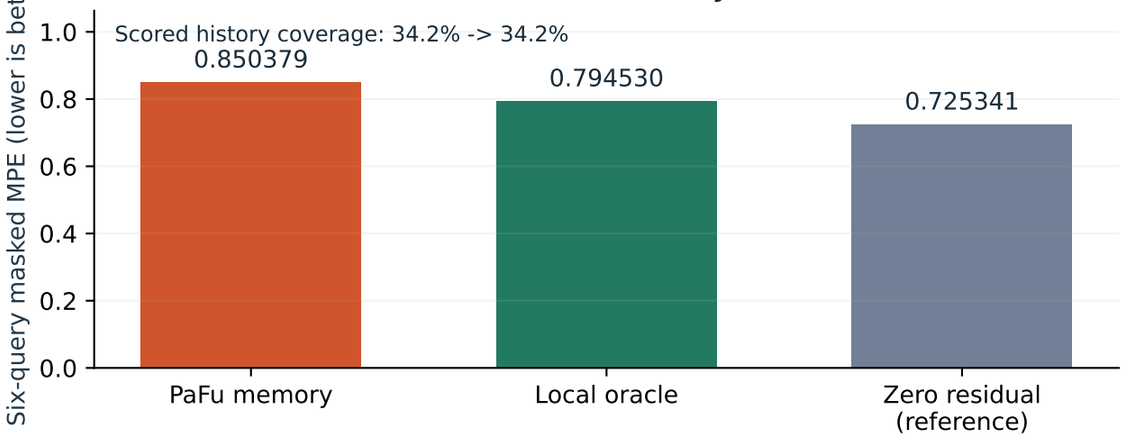
Deleting the high-attention entry helps mainly through signal; zero residual is still better than oracle.
Previously selected case reused; 129/864 parents have positive local regret, 81/129 improve signal.
Archived residual accounting, not RGB. Signal is pre-injection slow-model residual, not physical ground truth.

02 | Lower oracle loss can sacrifice signal
EXACT / source 24 / K64/PaFu / t=528

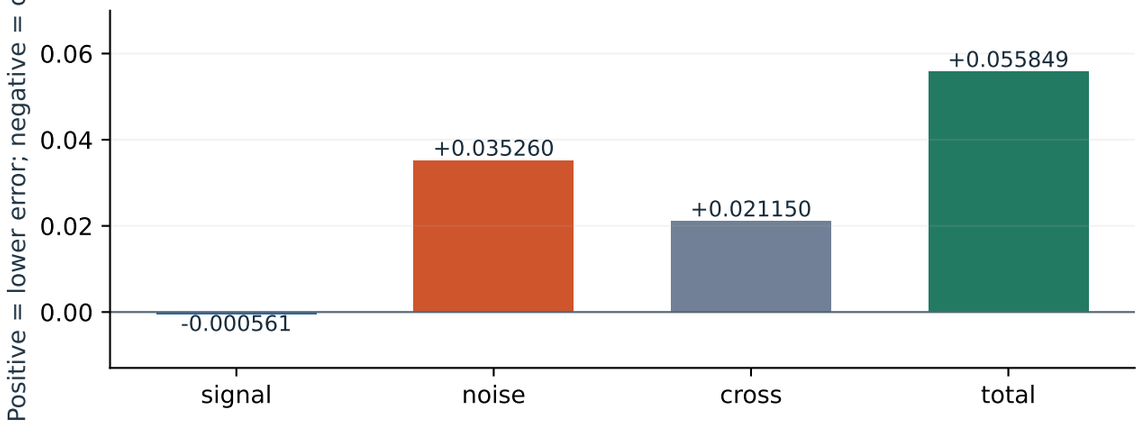
A. Observed candidate history and archived future-attention labels



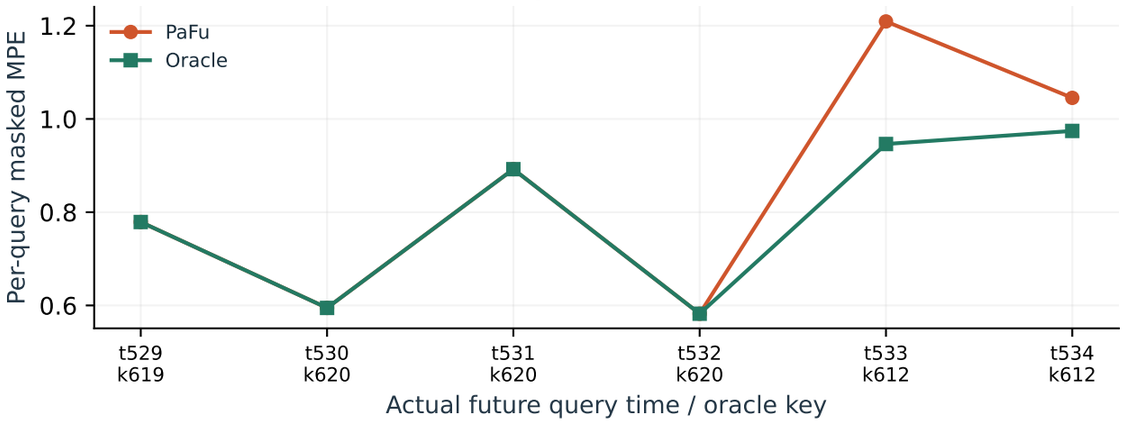
B. Same K / same reader / same future objective



C. Benefit of oracle deletion: PaFu minus oracle



D. Which future queries change?



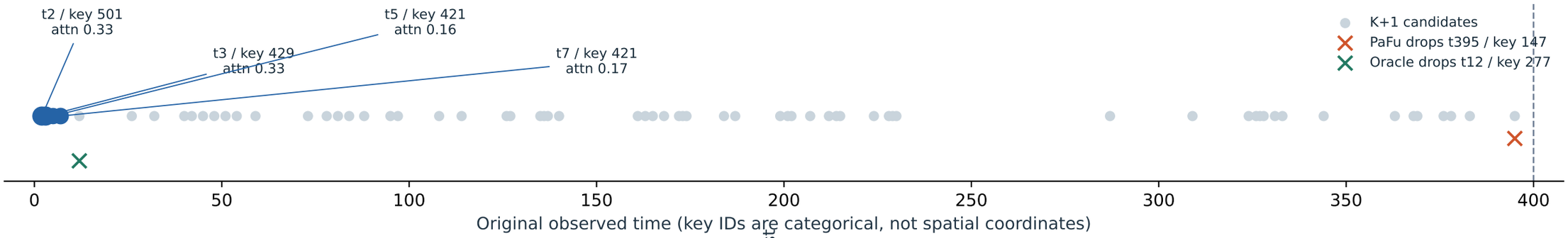
E. Covered scored coordinates / total scored

time / key	PaFu	Oracle
529 / 619	0/11	0/11
530 / 620	0/12	0/12
531 / 620	0/12	0/12
532 / 620	0/13	0/13
533 / 612	12/12	12/12
534 / 612	13/13	13/13

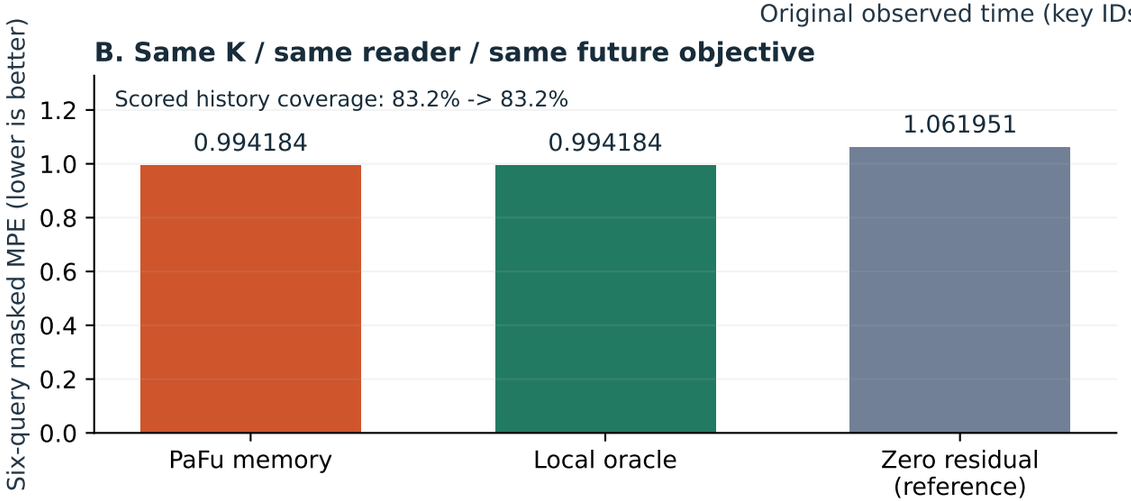
Coverage is unchanged: deleting one same-key record lowers realized noise but worsens signal.
Deterministic first qualifying K64/PaFu case: 24/96 qualify; all methods: 48/129 positive-regret parents worsen signal.
Archived residual accounting, not RGB. Signal is pre-injection slow-model residual, not physical ground truth.

03 | A positive control: memory is useful here
EXACT / source 24 / K64/PaFu / t=400

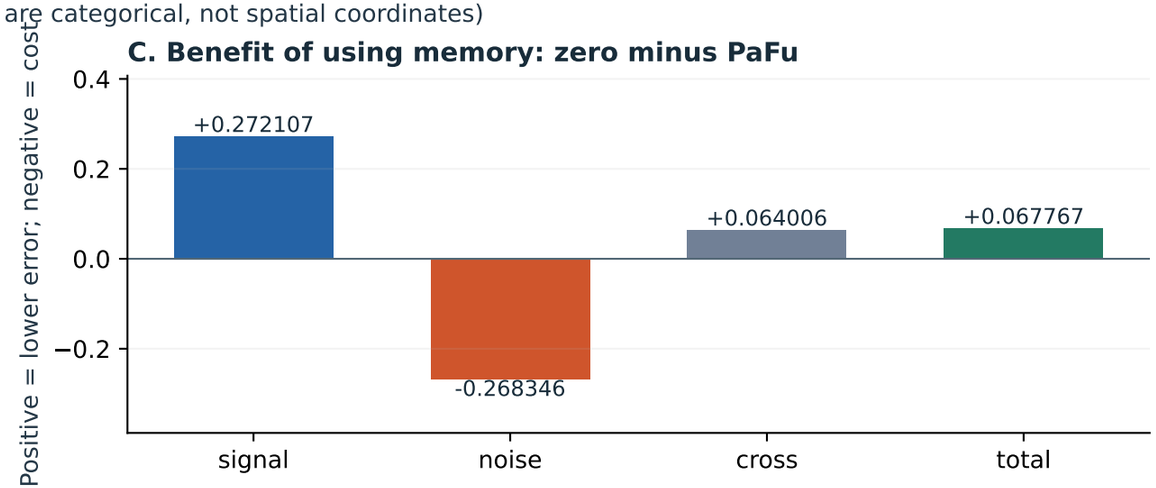
A. Observed candidate history and archived future-attention labels



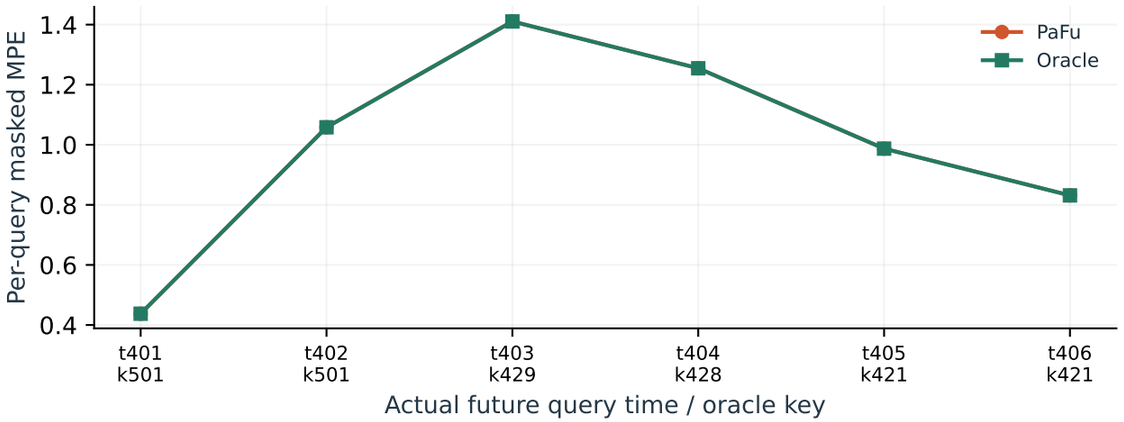
B. Same K / same reader / same future objective



C. Benefit of using memory: zero minus PaFu



D. Which future queries change?

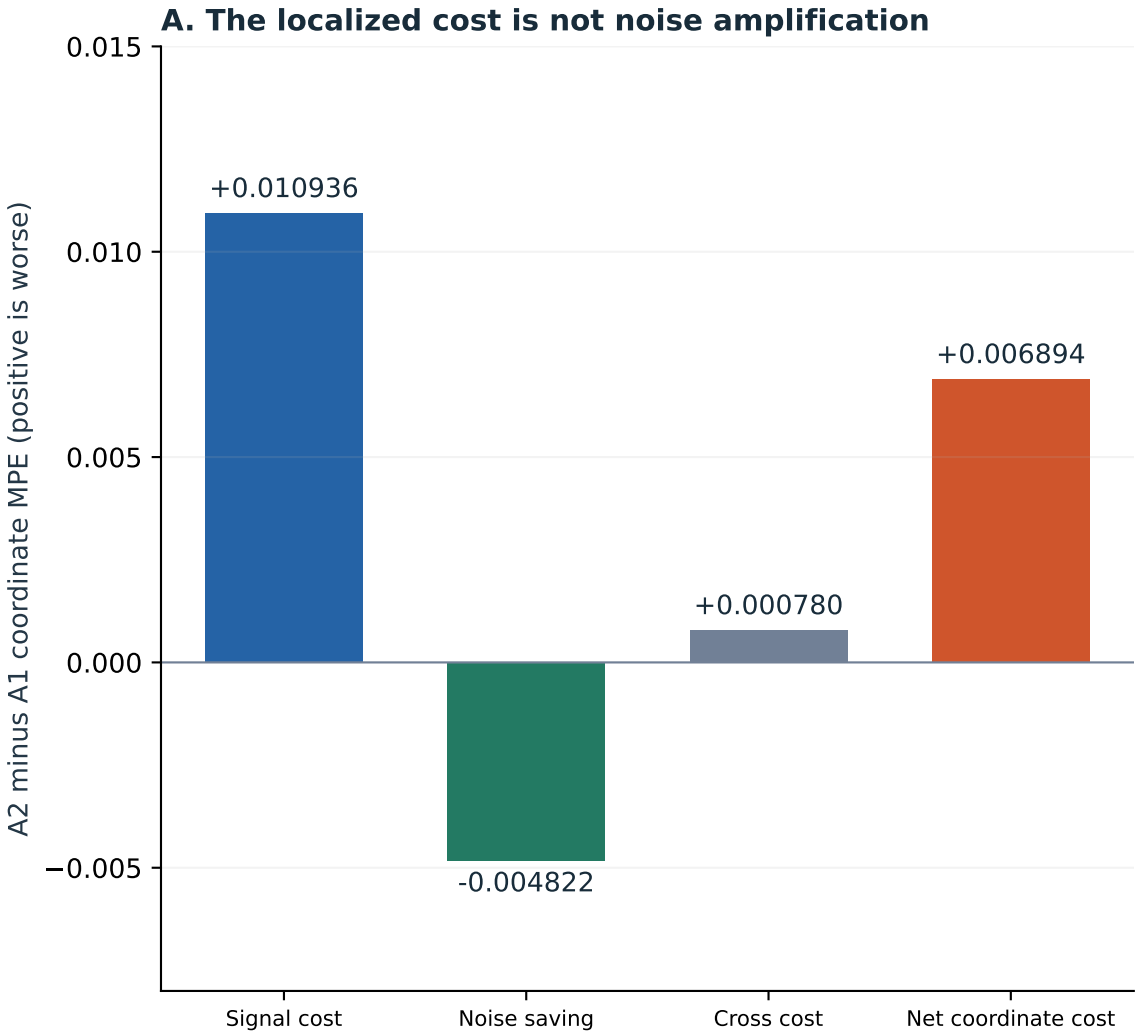


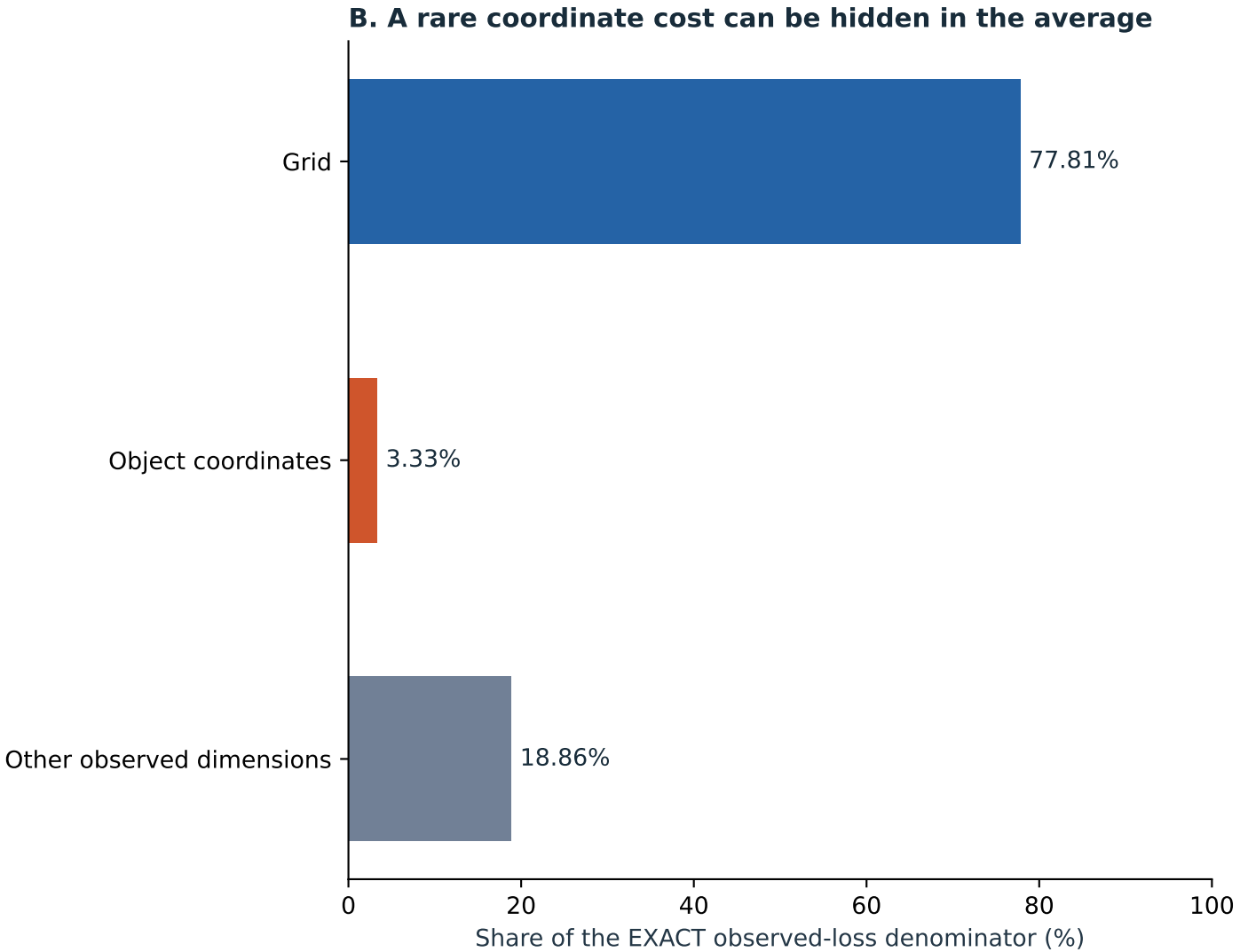
E. Covered scored coordinates / total scored

time / key	PaFu	Oracle
401 / 501	22/22	22/22
402 / 501	22/22	22/22
403 / 429	22/22	22/22
404 / 428	0/24	0/24
405 / 421	26/26	26/26
406 / 421	27/27	27/27

PaFu and local oracle tie; memory improves signal and total loss versus zero, despite a noise cost.
Deterministic first qualifying K64/PaFu case: 3/96 qualify; all methods: 25/864. This is one snapshot, not online superiority.
Archived residual accounting, not RGB. Signal is pre-injection slow-model residual, not physical ground truth.

04 | A2 trades signal fidelity for denoising
EXACT object coordinates: existing aggregate, not a reconstructed scene





CHANGE coordinate error improves; a general object-memory failure was not established.
Bookkeeping: fixing only this observed cost would change combined retention MPE by 0.0001157 (0.0125%).
Not a tested policy, attainable guarantee, or bound on all methods. Existing negative main-line decision is preserved.